

Constructors And Destructors

7.1 Introduction

Member variables can be initialised while creating the objects by using constructors. We can also destroy the objects when they are not required using destructors.

Classes have a very complicated structure. We can use constructors and destructors to initialise member variables of a class or destroy class objects. Besides, initialisations for objects construction also indicate memory allocation for the objects. Similarly, besides cleaning up objects, destructors de-allocation of memory used by the objects.

Constructors and destructors are usually declared within the declaration of a class. Here, you can define them either inline or external to the class declaration. Default arguments can be included in the constructors. Constructors and destructors also have some limitations.

Features of return values and return types are not found in constructors and destructors. Destructors don't take any argument. Programmers cannot use references and pointers in constructors. The keyword `virtual` cannot be used while declaring a constructor. Class objects that include constructors and destructors cannot be inserted in the unions. Constructors and destructors always maintain the access rules of member functions. The compiler where the whole program is run can automatically call constructors while defining class objects. Similarly, the compiler can automatically call destructors where the class objects turn to be insignificant.